

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Practical

Practical – II

Roll No. : T048	Name: Lavkush Vishwakarma
Class : TYIT	Date/Time of Submission : 30/06/2026

Aim: Core JavaScript Concepts for MERN.

1.A). Create functions and call them with parameters.

SOURCE CODE:-

```
function calculateSalary(salary, name) {  
    let bonus = 5000;  
    let totalSalary = salary + bonus;  
    console.log("Employee Name: " + name);  
    console.log("Base Salary: " + salary);  
    console.log("Bonus: " + bonus);  
    console.log("Total Salary: " + totalSalary);  
}  
calculateSalary(30000, "John Doe");
```

2nd

```
function calculate(product, quantity, price) {  
    let total = quantity * price;  
    let gst = total * 18 / 100;  
    let finalAmount = total + gst;  
  
    console.log("Product:", product);  
    console.log("Quantity:", quantity);  
    console.log("Price:", price);  
    console.log("Total:", total);  
    console.log("GST:", gst);  
    console.log("Final Amount:", finalAmount);  
}  
calculate("Book", 5, 100);  
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P2.js  
Employee Name: John Doe  
Base Salary: 30000  
Bonus: 5000  
Total Salary: 35000  
PS D:\Users\Lavkush\Study\MERN> node P3.js  
Product: Book  
Quantity: 5  
Price: 100  
Total: 500  
GST: 90  
Final Amount: 590  
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```

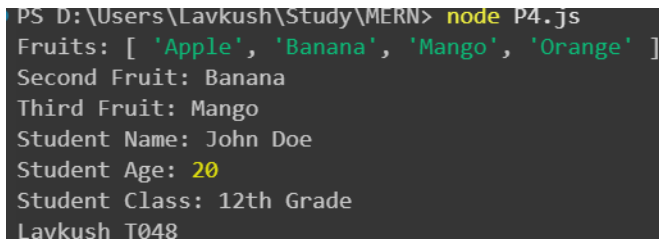
1.B). Work with arrays and objects in JavaScript.

SOURCE CODE:-

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```
let fruits = ["Apple", "Banana", "Mango", "Orange"];
let student = {
  name: "John Doe",
  age: 20,
  class: "12th Grade",
};
console.log("Fruits:", fruits);
console.log("Second Fruit:", fruits[1]);
console.log("Third Fruit:", fruits[2]);
console.log("Student Name:", student.name);
console.log("Student Age:", student.age);
console.log("Student Class:", student.class);
console.log("Lavkush T048");
```

OUTPUT:-



```
PS D:\Users\Lavkush\Study\MERN> node P4.js
Fruits: [ 'Apple', 'Banana', 'Mango', 'Orange' ]
Second Fruit: Banana
Third Fruit: Mango
Student Name: John Doe
Student Age: 20
Student Class: 12th Grade
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```

1.C). Demonstrate use of arrow functions.

SOURCE CODE:-

```
let add = (a, b) => {
  return a + b;
};
let sub = (a, b) => {
  return a - b;
};
let mul = (a, b) => {
  return a * b;
};
let div = (a, b) => {
  return a / b;
};
console.log("Addition: ", add(34, 96));
console.log("Subtraction: ", sub(34, 96));
console.log("Multiplication: ", mul(34, 96));
console.log("Division: ", div(34, 96));
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P5.js
Addition: 130
Subtraction: -62
Multiplication: 3264
Division: 0.3541666666666667
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```

1.D). Write a program using array methods (map, filter, reduce).

SOURCE CODE:-

```
let numbers = [10,20,30,40,50]
let doubled = numbers.map(num => num * 2);
let greater25 = numbers.filter(num => num > 25);
let total = numbers.reduce((sum, num) => sum + num, 0);
console.log("Original Array:",numbers);
console.log("Map:",doubled);
console.log("Filter:",greater25);
console.log("Reduce:",total);
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P6.js
Original Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
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```

1.E). Create a program that manipulates objects and displays output.

SOURCE CODE:-

```
let student = {
  name: "Zeus",
  rollno: 14,
  age: 19,
  marks: 95
};
console.log("Original Object:");
console.log(student);
//Accessing Properties
console.log("Accessing Properties")
console.log("Name:",student.name);
console.log("Marks:",student.marks);
//Updating A Property
student.marks = 99;
console.log("After updating")
console.log(student);
//Adding A New Property
student.city = "Mumbai";
console.log("After Adding A New Property ")
console.log(student);
//Delete A Property
delete student.rollno;
```

```
console.log("After Deleting A Property ")
console.log(student);
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P7.js
Original Object:
{ name: 'Zeus', rollno: 14, age: 19, marks: 95 }
Accessing Properties
Name: Zeus
Marks: 95
After updating
{ name: 'Zeus', rollno: 14, age: 19, marks: 99 }
After Adding A New Property
{ name: 'Zeus', rollno: 14, age: 19, marks: 99, city: 'Mumbai' }
After Deleting A Property
{ name: 'Zeus', age: 19, marks: 99, city: 'Mumbai' }
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```

1.F). Write programs using loops and template literals.

SOURCE CODE:-

```
let students = [
  {
    name: "Zeus",
    rollno: 14,
    age: 19,
    marks: 95
  },
  {
    name: "Bhavesh",
    rollno: 39,
    age: 19,
    marks: 91
  },
  {
    name: "Lavkush",
    rollno: 48,
    age: 19,
    marks: 94
  }
];
function display(s) {
  console.log("Student Details:")
  for (let stud of s) {
    console.log(`Name : ${stud.name}`);
    console.log(`Age : ${stud.age}`);
    console.log(`Roll no : ${stud.rollno}`);
    console.log(`Marks : ${stud.marks}`);
  }
}
display(students)
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P8.js
Student Details:
Name : Zeus
Age : 19
Roll no : 14
Marks : 95
Name : Bhavesh
Age : 19
Roll no : 39
Marks : 91
Name : Lavkush
Age : 19
Roll no : 48
Marks : 94
Lavkush T048
```